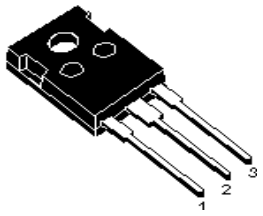
# STW26NM60

## N-CHANNEL 600V - 0.125Ω - 26A TO-247
## Zener-Protected MDmesh™Power MOSFET

| TYPE | $V_{DSS}$ | $R_{DS(on)}$ | $I_D$ |
|---|---|---|---|
| STW26NM60 | 600 V | < 0.135 Ω | 30 A |

- TYPICAL $R_{DS}(on)$ = 0.125Ω
- HIGH dv/dt AND AVALANCHE CAPABILITIES
- IMPROVED ESD CAPABILITY
- LOW INPUT CAPACITANCE AND GATE CHARGE
- LOW GATE INPUT RESISTANCE

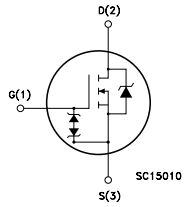



**TO-247**

### DESCRIPTION

The MDmesh™ is a new revolutionary MOSFET technology that associates the Multiple Drain process with the Company's PowerMESH™ horizontal layout. The resulting product has an outstanding low on-resistance, impressively high dv/dt and excellent avalanche characteristics. The adoption of the Company's proprietary strip technique yields overall dynamic performance that is significantly better than that of similar competition's products.

### INTERNAL SCHEMATIC DIAGRAM




### APPLICATIONS

The MDmesh™ family is very suitable for increasing power density of high voltage converters allowing system miniaturization and higher efficiencies.

### ORDERING INFORMATION

| SALES TYPE | MARKING | PACKAGE | PACKAGING |
|---|---|---|---|
| STW26NM60 | W26NM60 | TO-247 | TUBE |

## ABSOLUTE MAXIMUM RATINGS

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| $V_{DS}$ | Drain-source Voltage ($V_{GS}$ = 0) | 600 | V |
| $V_{DGR}$ | Drain-gate Voltage ($R_{GS}$ = 20 kΩ) | 600 | V |
| $V_{GS}$ | Gate- source Voltage | ± 30 | V |
| $I_D$ | Drain Current (continuous) at $T_C$ = 25°C | 30 | A |
| $I_D$ | Drain Current (continuous) at $T_C$ = 100°C | 18.9 | A |
| $I_{DM}$ (•) | Drain Current (pulsed) | 120 | A |
| $P_{TOT}$ | Total Dissipation at $T_C$ = 25°C | 313 | W |
| | Derating Factor | 2.5 | W/°C |
| $V_{ESD(G-S)}$ | Gate source ESD(HBM-C=100pF, R=1.5KΩ) | 6000 | V |
| dv/dt (1) | Peak Diode Recovery voltage slope | 15 | V/ns |
| $T_j$<br>$T_{stg}$ | Operating Junction Temperature<br>Storage Temperature | -55 to 150 | °C |

(•) Pulse width limited by safe operating area
(1) $I_{SD}$ ≤26A, di/dt ≤200A/µs, $V_{DD} \leq V_{(BR)DSS}$, $T_j \leq T_{JMAX.}$

## THERMAL DATA

| | | | |
|---|---|---|---|
| Rthj-case | Thermal Resistance Junction-case Max | 0.4 | °C/W |
| Rthj-amb | Thermal Resistance Junction-ambient Max | 62.5 | °C/W |
| $T_l$ | Maximum Lead Temperature For Soldering Purpose | 300 | °C |

## AVALANCHE CHARACTERISTICS

| Symbol | Parameter | Max Value | Unit |
|---|---|---|---|
| $I_{AR}$ | Avalanche Current, Repetitive or Not-Repetitive (pulse width limited by $T_j$ max) | 13 | A |
| $E_{AS}$ | Single Pulse Avalanche Energy (starting $T_j$ = 25 °C, $I_D = I_{AR}$, $V_{DD}$ = 50 V) | 740 | mJ |

## GATE-SOURCE ZENER DIODE

| Symbol | Parameter | Test Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $BV_{GSO}$ | Gate-Source Breakdown Voltage | Igss=± 1mA (Open Drain) | 30 | | | V |

## PROTECTION FEATURES OF GATE-TO-SOURCE ZENER DIODES

The built-in back-to-back Zener diodes have specifically been designed to enhance not only the device's ESD capability, but also to make them safely absorb possible voltage transients that may occasionally be applied from gate to source. In this respect the Zener voltage is appropriate to achieve an efficient and cost-effective intervention to protect the device's integrity. These integrated Zener diodes thus avoid the usage of external components.

**ELECTRICAL CHARACTERISTICS** ($T_{CASE}$ =25°C UNLESS OTHERWISE SPECIFIED)

ON/OFF

| Symbol | Parameter | Test Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $V_{(BR)DSS}$ | Drain-source Breakdown Voltage | $I_D$ = 250 µA, $V_{GS}$ = 0 | 600 | | | V |
| $I_{DSS}$ | Zero Gate Voltage Drain Current ($V_{GS}$ = 0) | $V_{DS}$ = Max Rating<br>$V_{DS}$ = Max Rating, $T_C$ = 125 °C | | | 10<br>100 | µA<br>µA |
| $I_{GSS}$ | Gate-body Leakage Current ($V_{DS}$ = 0) | $V_{GS}$ = ± 20V | | | ±10 | µA |
| $V_{GS(th)}$ | Gate Threshold Voltage | $V_{DS}$ = $V_{GS}$, $I_D$ = 250µA | 3 | 4 | 5 | V |
| $R_{DS(on)}$ | Static Drain-source On Resistance | $V_{GS}$ = 10V, $I_D$ = 13 A | | 0.125 | 0.135 | Ω |

DYNAMIC

| Symbol | Parameter | Test Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $g_{fs}$ (1) | Forward Transconductance | $V_{DS}$ = 15 V, $I_D$ = 13 A | | 20 | | S |
| $C_{iss}$<br>$C_{oss}$<br>$C_{rss}$ | Input Capacitance<br>Output Capacitance<br>Reverse Transfer Capacitance | $V_{DS}$ = 25V, f = 1 MHz, $V_{GS}$ = 0 | | 2900<br>900<br>40 | | pF<br>pF<br>pF |

SWITCHING ON

| Symbol | Parameter | Test Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $t_{d(on)}$<br>$t_r$ | Turn-on Delay Time<br>Rise Time | $V_{DD}$ = 300V, $I_D$ = 13 A<br>$R_G$ = 4.7Ω $V_{GS}$ = 10 V<br>(Resistive Load see, Figure 3) | | 35<br>22 | | ns<br>ns |
| $Q_g$<br>$Q_{gs}$<br>$Q_{gd}$ | Total Gate Charge<br>Gate-Source Charge<br>Gate-Drain Charge | $V_{DD}$ = 480V, $I_D$ = 26 A,<br>$V_{GS}$ = 10V | | 73<br>20<br>37 | 102 | nC<br>nC<br>nC |

SWITCHING OFF

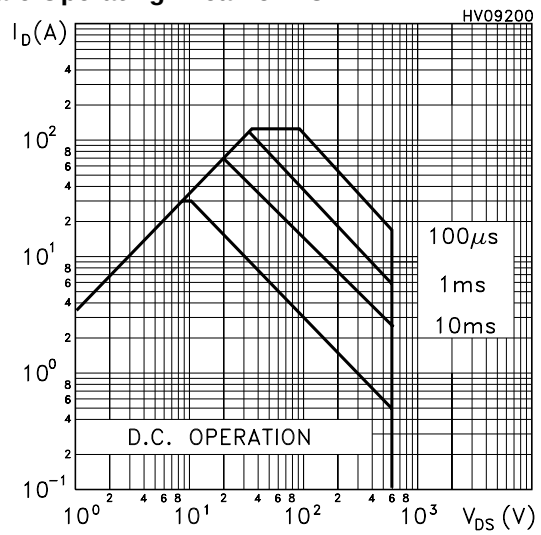
| Symbol | Parameter | Test Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $t_{r(Voff)}$<br>$t_f$<br>$t_c$ | Off-voltage Rise Time<br>Fall Time<br>Cross-over Time | $V_{DD}$ = 480V, $I_D$ = 26 A,<br>$R_G$ = 4.7Ω, $V_{GS}$ = 10V<br>(Inductive Load see, Figure 5) | | 14<br>20<br>40 | | ns<br>ns<br>ns |

SOURCE DRAIN DIODE

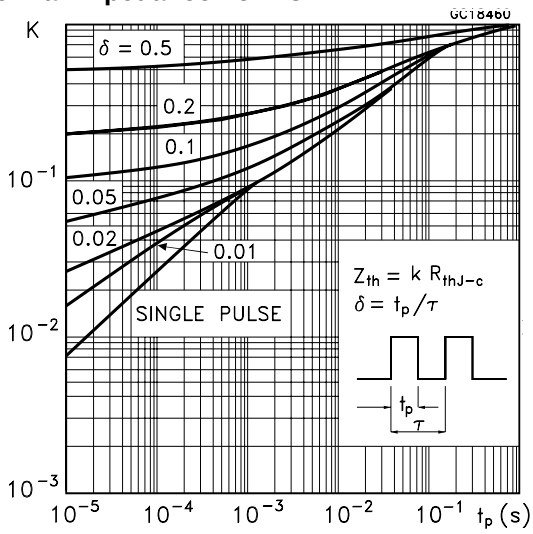
| Symbol | Parameter | Test Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $I_{SD}$<br>$I_{SDM}$ (2) | Source-drain Current<br>Source-drain Current (pulsed) | | | | 26<br>104 | A<br>A |
| $V_{SD}$ (1) | Forward On Voltage | $I_{SD}$ = 26 A, $V_{GS}$ = 0 | | | 1.5 | V |
| $t_{rr}$<br>$Q_{rr}$<br>$I_{RRM}$ | Reverse Recovery Time<br>Reverse Recovery Charge<br>Reverse Recovery Current | $I_{SD}$ = 26 A, di/dt = 100A/µs<br>$V_{DD}$ = 100 V, $T_j$ = 25°C<br>(see test circuit, Figure 5) | | 450<br>7<br>30.5 | | ns<br>µC<br>A |
| $t_{rr}$<br>$Q_{rr}$<br>$I_{RRM}$ | Reverse Recovery Time<br>Reverse Recovery Charge<br>Reverse Recovery Current | $I_{SD}$ = 26 A, di/dt = 100A/µs<br>$V_{DD}$ = 100 V, $T_j$ = 150°C<br>(see test circuit, Figure 5) | | 560<br>9<br>32.5 | | ns<br>µC<br>A |

Note: 1. Pulsed: Pulse duration = 300 µs, duty cycle 1.5 %.
2. Pulse width limited by safe operating area.

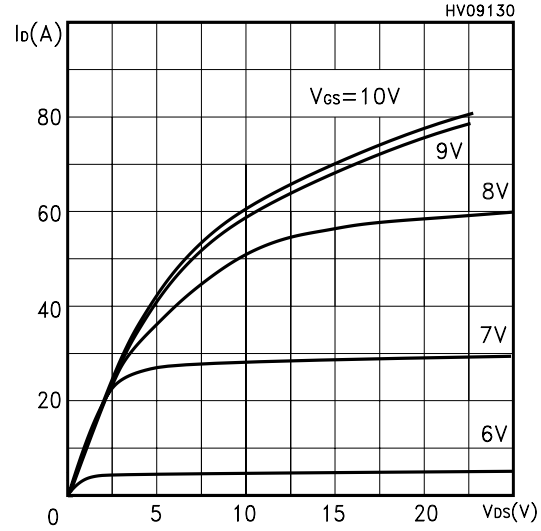
### Safe Operating Area For TO-247



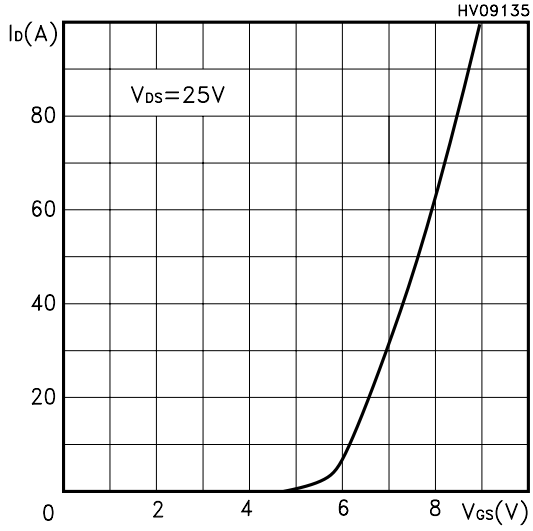

### Thermal Impedance For TO-247



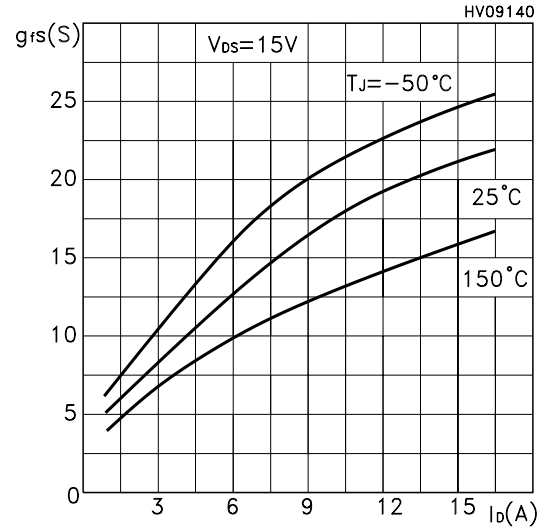

### Output Characteristics



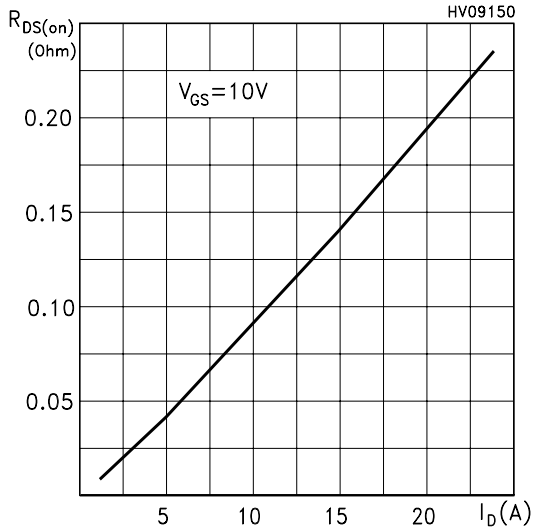

### Transfer Characteristics



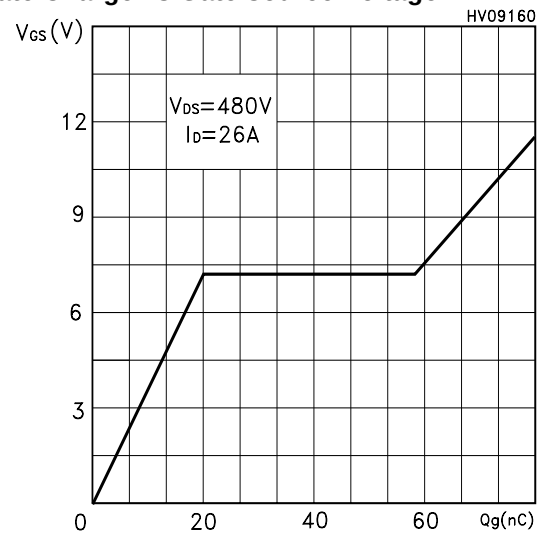

### Transconductance



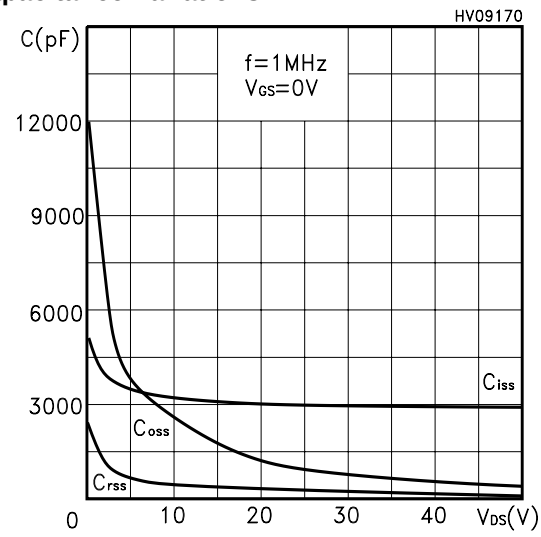

### Static Drain-source On Resistance



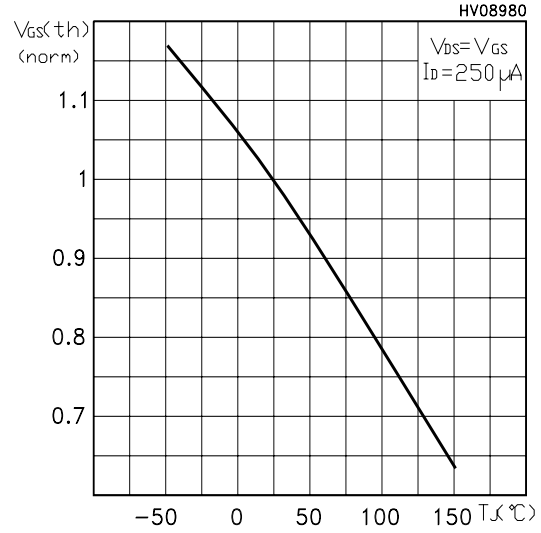
### Gate Charge vs Gate-source Voltage



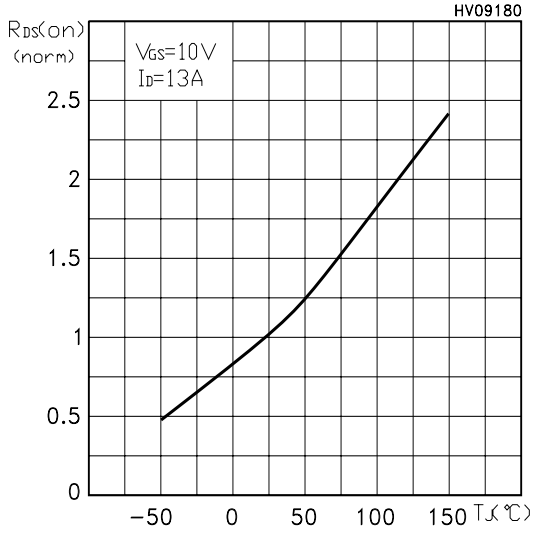

### Capacitance Variations




### Normalized Gate Threshold Voltage vs Temp.




### Normalized On Resistance vs Temperature




### Source-drain Diode Forward Characteristics

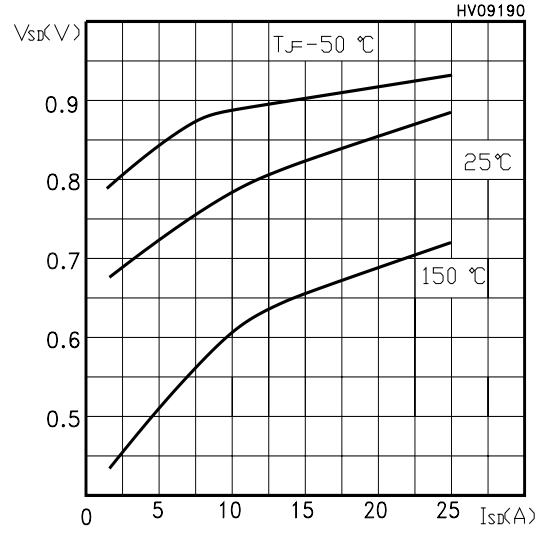


**Fig. 1:** Unclamped Inductive Load Test Circuit

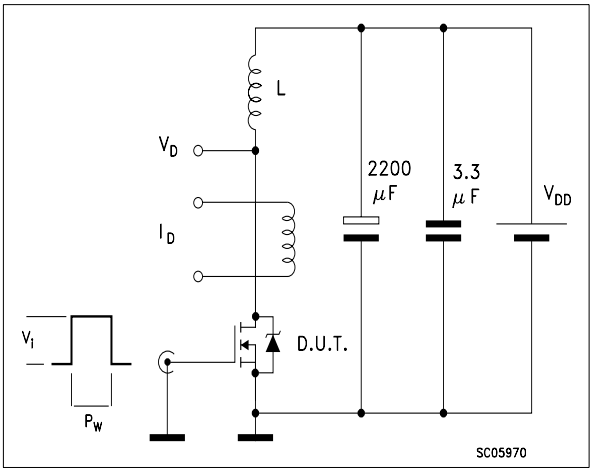



**Fig. 2:** Unclamped Inductive Waveform

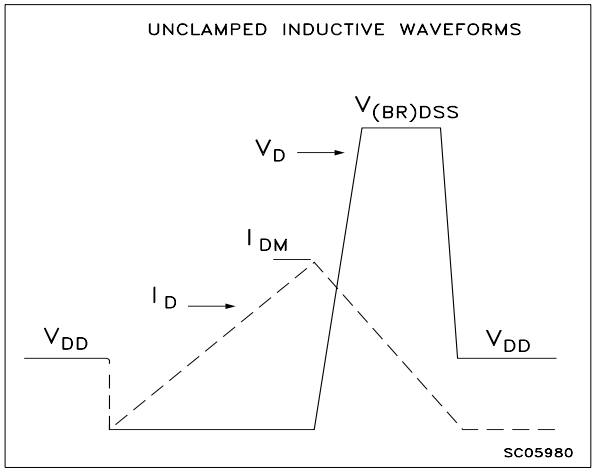



**Fig. 3:** Switching Times Test Circuit For Resistive Load

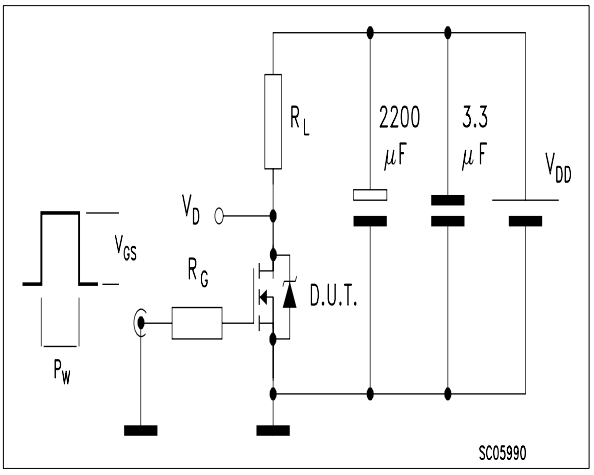



**Fig. 4:** Gate Charge test Circuit

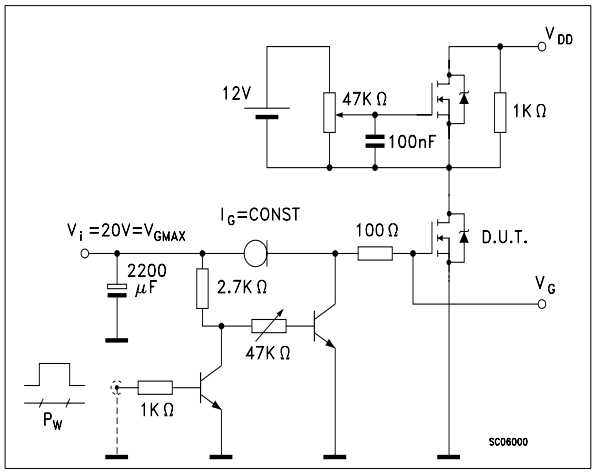
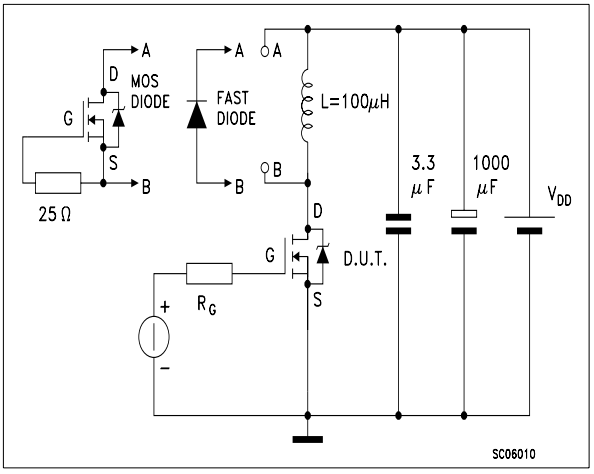
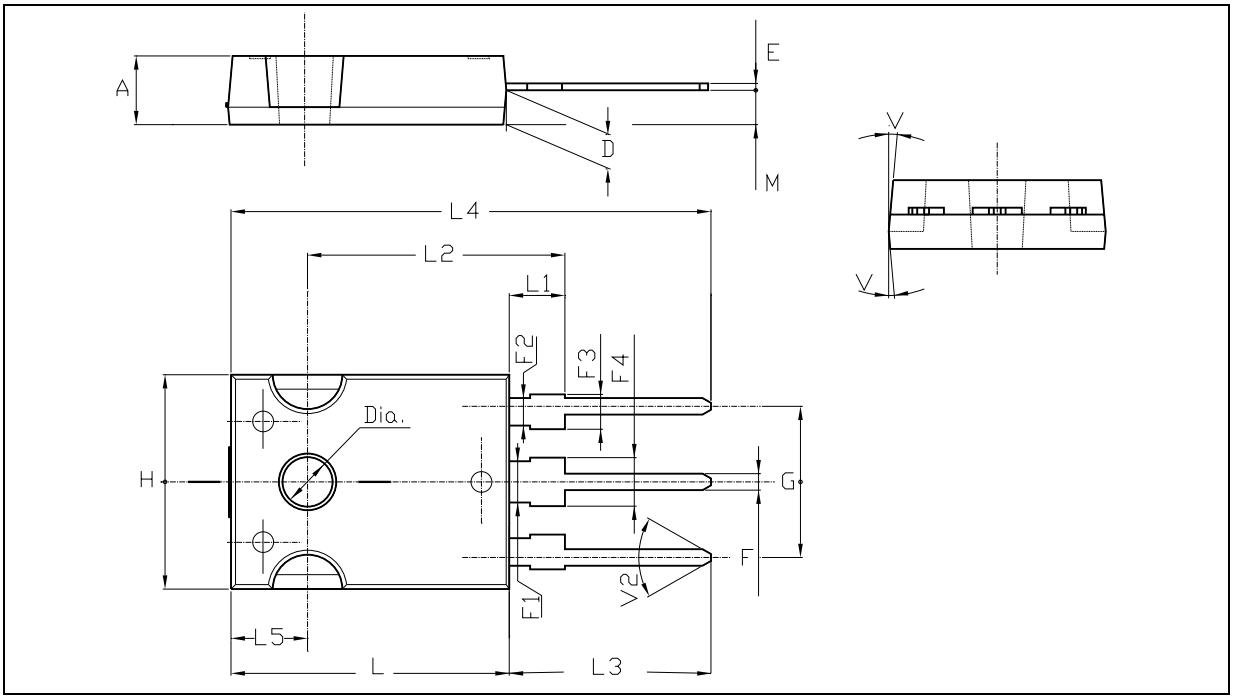



**Fig. 5:** Test Circuit For Inductive Load Switching And Diode Recovery Times



# TO-247 MECHANICAL DATA

| DIM. | mm. | | | inch | | |
|---|---|---|---|---|---|---|
| | MIN. | TYP | MAX. | MIN. | TYP. | MAX. |
| A | 4.85 | | 5.15 | 0.19 | | 0.20 |
| D | 2.20 | | 2.60 | 0.08 | | 0.10 |
| E | 0.40 | | 0.80 | 0.015 | | 0.03 |
| F | 1 | | 1.40 | 0.04 | | 0.05 |
| F1 | | 3 | | | 0.11 | |
| F2 | | 2 | | | 0.07 | |
| F3 | 2 | | 2.40 | 0.07 | | 0.09 |
| F4 | 3 | | 3.40 | 0.11 | | 0.13 |
| G | | 10.90 | | | 0.43 | |
| H | 15.45 | | 15.75 | 0.60 | | 0.62 |
| L | 19.85 | | 20.15 | 0.78 | | 0.79 |
| L1 | 3.70 | | 4.30 | 0.14 | | 0.17 |
| L2 | | 18.50 | | | 0.72 | |
| L3 | 14.20 | | 14.80 | 0.56 | | 0.58 |
| L4 | | 34.60 | | | 1.36 | |
| L5 | | 5.50 | | | 0.21 | |
| M | 2 | | 3 | 0.07 | | 0.11 |
| V | | 5º | | | 5º | |
| V2 | | 60º | | | 60º | |
| Dia | 3.55 | | 3.65 | 0.14 | | 0.143 |



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